



give me imp aws, gcp and azure services with what they are used for

Yes — here's a **practical cheat sheet** of the most important AWS, Azure, and GCP services, grouped by what you'll usually use them for in real projects. The service mappings below are widely recognized across the three clouds, such as EC2 vs Azure Virtual Machine vs Compute Engine, Lambda vs Azure Function vs Cloud Functions, and EKS/AKS/GKE for containers.^[1] ^[2] ^[3]

Compute

These are the core services you use to run apps, APIs, websites, and backend servers in the cloud. If you are coming from full-stack or DevOps work, this is usually the first group to learn.^[2] ^[1]

Use case	AWS	Azure	GCP
Virtual machine / server	EC2 (Elastic Compute) — run Linux or Windows VMs, host backend apps, custom servers, cron jobs, and full environments where you control the OS. ^[1]	Azure Virtual Machine — run VMs for web apps, enterprise apps, Windows workloads, or custom infrastructure. ^[1]	Google Compute Engine — run virtual machines for websites, APIs, data processing, and general-purpose compute. ^[1]
Platform as a Service	Elastic Beanstalk — deploy web apps without managing low-level infrastructure manually. ^[1]	App Service — host web apps, REST APIs, and backend services with managed deployment. ^[1]	Google App Engine — deploy applications on a managed platform without handling servers directly. ^[1]
Serverless functions	Lambda — run backend code on events, HTTP triggers, queues, automation, and microservices without managing servers. ^[1]	Azure Function — run event-driven code for APIs, automations, background jobs, and integrations. ^[1]	Cloud Functions — execute small event-based functions for APIs, triggers, and backend automation. ^[1]
Containers / Kubernetes	Elastic Kubernetes Service (EKS) or Elastic Container Service — deploy Docker containers and microservices at scale. ^[1]	Azure Kubernetes Service (AKS) — run and manage Kubernetes clusters for containerized applications. ^[1]	Google Kubernetes Engine (GKE) — run Kubernetes workloads with strong support for containerized apps. ^[1] ^[4]

Storage and data

These services are used to store files, application data, and analytics data. In most real apps, object storage plus one database service is the basic combination. [\[3\]](#) [\[2\]](#)

Use case	AWS	Azure	GCP
Object/file storage	S3 — store images, videos, backups, static websites, logs, and app files durably. [3]	Blob Storage — store unstructured data such as documents, media, and backup files. [3]	Cloud Storage — store files, datasets, backups, and data lake content. [2] [3]
Data lake	Lake Formation — build and manage a data lake for analytics and large-scale data processing. [2]	Data Share — share and organize large datasets for analytics workflows. [2]	Cloud Storage — often used as the base storage layer for data lakes. [2]
Data warehouse / analytics	Redshift — analyze large structured datasets and business intelligence workloads. [5]	Azure Synapse Analytics — run analytics and large-scale data warehousing. [5]	BigQuery — perform large-scale SQL analytics and data warehouse queries. [5] [3]
ETL / data pipelines	Glue and Kinesis Data Firehose — prepare, transform, and move data into analytics systems. [2]	Data Factory — build ETL pipelines and orchestrate data movement. [2]	Data Fusion, Dataflow, and Dataproc — process, transform, and move data for analytics and big data workloads. [2]

Networking and delivery

These services make your application reachable, fast, and reliable for users. They are essential when you want domain names, APIs, traffic distribution, and CDN acceleration. [\[2\]](#) [\[3\]](#)

Use case	AWS	Azure	GCP
DNS	Route 53 — manage domain names and route users to your application endpoints. [2]	DNS — host and manage DNS records for applications and services. [2]	Cloud DNS — provide managed DNS for routing users to services. [2]
Load balancing	Elastic Load Balancing (ELB) — distribute traffic across multiple servers or services. [2]	Application Gateway, Load Balancer, and Traffic Manager — balance and route traffic for apps and services. [2]	Cloud Load Balancing — distribute user traffic across instances and regions. [2]
CDN	CloudFront — cache and deliver static and dynamic content faster worldwide. [2]	Content Delivery Network — improve content delivery speed for users in different locations. [2]	Cloud CDN and Media CDN — accelerate delivery of websites, APIs, and media content. [2]
API management	API Gate way — create, publish, and manage APIs. [2]	API Apps and API Management — build and manage APIs for applications and integrations. [2]	Apigee API Management — design, secure, and operate APIs. [2]

Security, monitoring, and AI

These are the services you use for user access, observability, security posture, and machine learning features. They become important very quickly once your app has real users, production traffic, or AI requirements.^[4] ^[2]

Use case	AWS	Azure	GCP
Identity and access	IAM — control users, roles, and permissions for cloud resources. ^[2]	Entra ID — handle identity, authentication, and access management. ^[2]	Cloud Identity — manage users and access across Google Cloud resources. ^[2]
Monitoring	CloudWatch — collect logs, metrics, and alerts for apps and infrastructure. ^[2]	Monitor and Anomaly Detector — observe health, metrics, and unusual behavior. ^[2]	Operations and Network Intelligence Center — monitor workloads and network behavior. ^[2]
Security posture	Security Hub — centralize security findings and improve cloud security visibility. ^[2]	Security — protect cloud resources and improve security management. ^[2]	Security Command Center — view and manage cloud security risks centrally. ^[2] ^[6]
Machine learning	SageMaker — train, validate, and deploy ML models. ^[2]	Machine Learning — build and deploy ML workflows and models. ^[2]	Vertex AI — train, manage, and deploy machine learning models. ^[2]
NLP / text AI	Comprehend — analyze text for NLP tasks. ^[2]	Text Analytics — process and understand text data. ^[2]	Natural Language AI — perform NLP tasks on text. ^[2]
Vision / OCR / speech	Rekognition, Textract, Transcribe — handle image analysis, document text extraction, and speech transcription. ^[2]	Cognitive Services for Vision, Form Recognizer, and Cognitive Services for Speech to Text — support computer vision, OCR, and speech tasks. ^[2]	Vision AI, Document AI, and Speech-to-Text — provide image, document, and audio intelligence services. ^[2]

What to learn first

If your goal is interviews, internships, or project building, start with the services that appear across all three providers in similar roles: virtual machines, object storage, serverless functions, Kubernetes/container services, IAM, load balancers, CDN, monitoring, and one ML platform. The sources also show that AWS has the largest overall catalog, Azure is strong for Microsoft and hybrid environments, and GCP is especially strong for containers and data-heavy workloads.^[7] ^[1] ^[4]

- Learn this cross-cloud order first: VM service, object storage, IAM, serverless, database or warehouse, load balancer, DNS, monitoring, Kubernetes.^[1] ^[2]
- For AWS, focus first on EC2, S3, IAM, Lambda, EKS, CloudFront, Route 53, CloudWatch, and SageMaker.^[3] ^[1] ^[2]
- For Azure, focus first on Azure Virtual Machine, App Service, Azure Function, AKS, Blob Storage, Entra ID, Monitor, API Management, and Machine Learning.^[1] ^[2]

- For GCP, focus first on Compute Engine, App Engine, Cloud Functions, GKE, Cloud Storage, Cloud DNS, Cloud Load Balancing, Security Command Center, BigQuery, and Vertex AI.^{[2] [3] [1]}

Would you like me to turn this into a 1-page interview revision table with only the top 25 services and easy memory tricks?

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1. <https://www.coursera.org/in/articles/aws-vs-azure-vs-google-cloud>
2. <https://www.datacamp.com/cheat-sheet/aws-azure-and-gcp-service-comparison-for-data-science-and-a>
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3. <https://docs.cloud.google.com/docs/get-started/aws-azure-gcp-service-comparison>
4. <https://www.bmc.com/blogs/aws-vs-azure-vs-google-cloud-platforms/>
5. <https://github.com/in28minutes/learning-paths-cloud-and-devops/blob/main/aws-vs-azure-vs-gcp.md>
6. <https://www.future-processing.com/blog/cloud-services-comparison-market-share-main-differences/>
7. <https://www.digitalocean.com/resources/articles/comparing-aws-azure-gcp>
8. <https://www.geeksforgeeks.org/devops/aws-vs-google-cloud-platform-vs-azure/>
9. <https://www.varonis.com/blog/aws-vs-azure-vs-google>
10. <https://cast.ai/blog/cloud-pricing-comparison/>
11. <https://www.vcomsolutions.com/blog/comparing-cloud-platforms-aws-azure-and-gcp>
12. <https://www.datacamp.com/blog/aws-vs-azure-vs-gcp>
13. <https://www.megaport.com/blog/aws-azure-google-cloud-the-big-three-compared/>
14. <https://www.eccouncil.org/cybersecurity-exchange/cloud-security/biggest-cloud-service-providers/>
15. <https://learn.microsoft.com/en-us/azure/architecture/aws-professional/>